

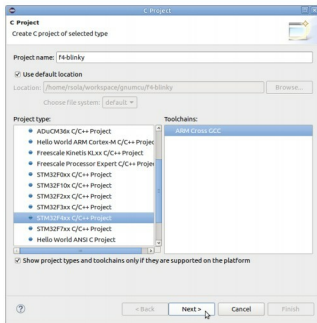


Figure 1: Project template selection

parameters like Flash memory size, clock speed, etc. Choose the Blinky template under *Content* type, and leave other options with default values for the initial work as shown in Figure 2. Follow default values for *Folder* settings and select configurations in the next two wizards. In the final wizard, provide the path of the *bin* directory under the extracted toolchain and hit *Finish* to complete the set-up process.

Right-click on the generated project, i.e., *f4-blinky* under the *Project Explorer* pane, and then click on *Build Project*. On successfully completing the build, you can see the following console output:

Finished building target: f4-blinky.elf

Invoking: GNU ARM Cross Create Flash Image
arm-none-eabi-objcopy -O ihex "f4-blinky.elf" "f4-blinky.hex"
Finished building: f4-blinky.hex

Invoking: GNU ARM Cross Print Size
arm-none-eabi-size --format=berkeley "f4-blinky.elf"
text data bss dec hex filename
8599 160 420 9179 23db f4-blinky.elf
Finished building: f4-blinky.siz

Project settings

Once the project is generated, you can tune various options for the preprocessor, C/C++ compiler, linker, etc. For this, right-click on the project, and go to *Properties* --> *C/C++ Build* --> *Settings*.

Under the *Tool Settings* tab, you can choose various options like Target, Assembler, Compiler, Linker, Flash,

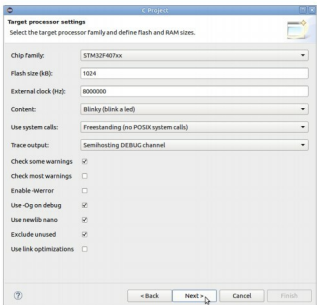


Figure 2: Target configuration

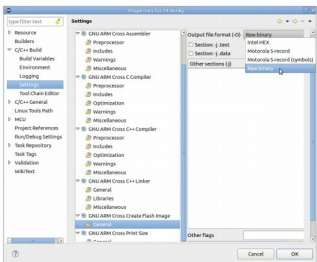


Figure 3: Project settings

etc. Plugins will generate suitable options of the toolchain for each selection. For example, if you want to generate a Flash image in Raw Binary format instead of Intel HEX, choose the output file format under *GNU ARM Cross Create Flash Image* --> *Output File Format*. Likewise, you can play around with various options under this tab view and check the selected values under 'All Options' expected for various tools in CLI mode. This way you can also learn about the various options to be used with a toolchain in simple interactive steps.

Flashing the code – Run/Debug

Once project settings are customised, rebuild the project. Right click the project and navigate to *Debug* --> *Debug As* --> *Debug Configurations*. Double-click on *GDB*